

Black Holes from Hyperbits and from FFGFT

Comparison of two algebraic-geometric approaches — focus: Landauer's principle, bit values, and system-dependent bit energy

Johann Pascher
ORCID 0009-0000-6518-4064

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Legend

[K] core result **[B]** algebraically proven **[S]** sketch **[Q]** source **[X]** excluded
[POSTULATE] axiom

Abstract

In “Black Holes from Hyperbits” (IPI 2026) Douglas Matzke derives the entire physics of black holes from the Clifford-algebra axiom $e_i^2 = +1$, without general relativity and without free parameters. Central to this: the mass of a bit follows from Landauer's principle ($E_{\text{bit}} = k_B T_P \ln 2$) plus $E = mc^2$ and is therefore a *universal* number — $m_{\text{bit}} = m_P \ln 2 / (2\pi) \approx 0.11 m_P$, independent of the physical system.

FFGFT (Fundamental Fractal Geometric Field Theory) arrives at a fundamentally different insight: bit values are *system-dependent*. The minimal information unit is the winding quantum $\Delta w = 1$ on T^4 (Doc. 257), and its energy is $E_{\text{bit}} = \hbar c / L$ — dependent on the physical scale L of the system. Landauer's $k_B T \ln 2$ is the special case of one particular scale and temperature, not a universal law (A271, A272, A273, Doc. 302).

This document compares both frameworks and works out the difference in the Landauer interpretation as theoretically essential.

.1 Why Matzke's algebra looks so simple

Before the technical comparisons, a narrative placement is worthwhile: where does the aesthetic appeal of the hyperbit approach come from – and where is it paid for? The two belong together.

The starting point: a sign

Matzke begins with $e_i^2 = +1$. That is a *bit* – something whose square is one, hence something with two states. No space, no time, no metric. Only the statement: there are distinguishable things.

The real trick is that the Clifford algebra immediately starts working on this. Take several such e_i , let them anticommute, and out of nothing arises a structure with dimension, orientation and rotations. Nothing was put in except “there are bits, and they are independent of one another” – and out comes geometry. That is the first reason for the beauty: **geometry as bookkeeping over distinguishability**, not as a presupposition.

Why $Cl(6)$ in particular

The second beautiful step: $Cl(6)$ decomposes into exactly three Witt pairs – three packets, each with one creator and one annihilator. Following Furey, this is precisely the structure of one fermion generation: three colours, one lepton. The three is not chosen, it *falls out*. Anyone who has tried to explain why there are three colours knows how satisfying that is.

The black-hole part

Now the actual coup. Bekenstein says: a black hole has entropy proportional to *area*, one bit per four Planck areas. Matzke turns this around: if a bit is an area, then a collection of bits is a volume – and the question “at what point does this collapse?” becomes a question about numbers, not about field equations. He sets the Compton wavelength equal to the Schwarzschild radius (the classical point where quantum mechanics and gravity meet) and obtains the Schwarzschild radius – **without ever using the Einstein equations**. For $20 M_\odot$: 59.1 km, exactly right.

And then the number that carries the appeal: the stability threshold sits at 6.41 bits. $Cl(6)$ has 6.000 bits. The universe consists of fermions sitting exactly 0.41 bits below the collapse threshold. Matter is stable because the smallest possible fermion algebra just barely fails to collapse. That is a fine story – terse, dramatic, with a concrete number.

Where the elegance is paid for

This 6.41 hangs on a postulate. To get from bits to masses, Matzke needs an exchange rate – what does a bit weigh? He takes Landauer ($E = k_B T \ln 2$) and evaluates it at the Planck temperature. Out comes $m_{\text{bit}} = m_P \ln 2 / (2\pi)$, a *universal* bit mass.

However: Landauer is a statement about thermal carrier ensembles, not about abstract information (A271–A273). And the Planck temperature is not the temperature of any physical system – it is an energy scale. The choice $T = T_P$ is a postulate that does not justify itself. The number 6.41 is therefore exactly as universal as that postulate – and Doc. 325 shows that the Hawking temperature does entirely without it.

The shortest version

Matzke’s algebra is beautiful because it shows how far pure combinatorics can go: from “there are bits” follow dimension, three colours, one fermion generation, the Schwarzschild radius. That is genuine reduction, not relabelling.

It is *simple* because it stops asking at three places: Bekenstein, Compton = Schwarzschild and Landauer at T_P are posited as bridges, not derived. Three borrowed bridges to a structure that is internally very frugal.

The contrast with FFGFT lies exactly there: FFGFT pays the price up front (deriving ξ , L_0 , justifying the torus) and afterwards needs no bridges – but the entry is harder. Matzke pays almost nothing up front and borrows three times later. Both accounts balance; they simply look elegant at different places.

.2 Starting points

Matzke

Single axiom: $e_i^2 = +1$ (hyperbit). **[POSTULATE]**

Physical units via three external bridges: Bekenstein ($4\ell_P^2$ per bit), Compton–Schwarzschild (ℓ_P as length scale), Landauer–Einstein ($m_{\text{bit}} = m_P \ln 2 / (2\pi)$).

FFGFT

Single parameter: $\xi = 4/30000$, derivable as smallest eigenvalue of the spectral operator $\hat{F}$ on $T^4/\mathbb{Z}_3$ **[K]** (Doc. 322). From ξ follow G , ℓ_P , $L_0 = \xi \cdot \ell_P$, $\alpha^{-1} = 137.036$, all lepton masses — without bridge axioms **[K]** (Docs. 180, 011, 006).

Minimal information unit: winding quantum $\Delta w = 1$, energy $E_{\text{bit}} = \hbar c / L$ (scale-dependent) (Doc. 257, A271).

.3 The Landauer question: universal vs. system-dependent bit values

This is the theoretically deepest difference between the two frameworks.

Matzke's position: a universal bit value

Matzke fixes the Landauer energy scale at the Planck temperature T_P :

$$E_{\text{bit}} = k_B T_P \ln 2 \quad \Rightarrow \quad m_{\text{bit}} = \frac{m_P \ln 2}{2\pi} \approx 0.110 m_P \quad (\text{Matzke, Ch. 6})$$

This value is *universal*: every bit carries the same mass, independent of the system (electron, black hole, vacuum). From it follow the stability threshold $n_{\text{thresh}} = 2\pi / (\sqrt{2} \ln 2) = 6.41$ and the Schwarzschild relation directly.

FFGFT position: system-dependent bit values

FFGFT understands Landauer's principle as a statement about *phase-space regions* of a thermal carrier ensemble, not about abstract information (A271):

*Counting proceeds by distinguishability (orthogonality), not by energy. Thermodynamics counts regions and does not read them — there is no thermodynamic truth detector. Sufficient for heat production is the **non-bijectivity of the region map**, not logical irreversibility.*

The fundamental bit energy in FFGFT is (Docs. 257, 302):

$$E_{\text{bit}} = \frac{\hbar c}{L} \quad (\text{FFGFT, Doc. 257})$$

where L is the physical scale of the system. Landauer's $k_B T \ln 2$ is the limiting case at one particular scale and bath temperature (A273: "two floors — the thermal $k_B T$ and the quantum-geometric $\hbar c/L$; the larger one governs").

The bit energy accordingly varies between scales:

System	Scale L	$E_{\text{bit}} = \hbar c/L$
Planck scale	$\ell_P = 1.62 \times 10^{-35} \text{ m}$	$m_P c^2 = 1.22 \times 10^{19} \text{ GeV}$
FFGFT floor	$L_0 = \xi \cdot \ell_P = 2.15 \times 10^{-38} \text{ m}$	$E_0/\xi \approx 56 \text{ PeV}$
Characteristic	$L_e = \hbar c/E_0 \approx 26.85 \text{ fm}$	$E_0 = 7.35 \text{ MeV}$
Nuclear physics	$\sim 1 \text{ fm}$	$\sim 197 \text{ MeV}$
Room-temperature bit	$L_T \sim k_B T/(\hbar c) \sim \mu\text{m}$	$\sim 3 \times 10^{-21} \text{ J} = 18 \text{ meV}$

FFGFT critique of Matzke's Landauer bridge

Matzke sets $T = T_P$ in $E_{\text{bit}} = k_B T \ln 2$. This yields a *universal* bit value. FFGFT objects: Landauer's bound holds for a thermal equilibrium ensemble at bath temperature T — a statement about the *carrier*, not about abstract information (A272). The Planck temperature T_P is not the temperature of the physical system (electron, proton, black hole in the cosmos); it is merely an energy scale. Landauer holds because of mechanics (Liouville), not because of a privileged scale.

Consequence: Matzke's universal stability threshold $n_{\text{thresh}} = 6.41$ hinges essentially on the choice $T = T_P$. With a system-dependent bit energy, n_{thresh} would itself be system-dependent. **[S]**

Two floors, one principle

FFGFT distinguishes (A273, Doc. 290):

Static Bit $\equiv$ winding quantum $\Delta_w = 1$ on T^4 : scale-universal, energy-independent as a counting quantity; the energy $E_{\text{bit}} = \hbar c/L$ is scale-specific.

Thermodynamic Irreversible erasure: $Q \geq k_B T \ln 2$ — cost in joules, dependent on the bath temperature T . This price has nothing to do with the information-theoretic quantity (A272: "no energy can be assigned to an information bit, only to its carrier").

Kinematic Reversible bit-flip time: $\tilde{T} = \hbar/E_{\text{bit}}$; Margolus–Levitin $\tau \geq \pi\hbar/(2E)$; energy sets the clock without loss (Doc. 290).

The four guises of a bit energy at the characteristic scale $E_0 = 7.35 \text{ MeV}$:

$$m_0 c^2 = E_0 = k_B T_0 = \hbar/\tilde{T} \quad \Rightarrow \quad T_0 = E_0/k_B \approx 8.5 \times 10^{10} \text{ K} \quad (1)$$

Landauer at room temperature gives $Q \approx 3 \times 10^{-21} \text{ J} \approx 18 \text{ meV}$ — eight orders of magnitude below E_0 . The statements "what the bit is" ($E_0 = 7.35 \text{ MeV}$) and "what erasure costs" ($k_B T \ln 2 = 18 \text{ meV}$ at 300 K) belong to different layers (Doc. 290, A271).

.4 Bekenstein entropy and Planck voxels

Matzke

1 bit = $\text{Cl}(6) = 3$ Witt pairs = 1 Planck voxel $4\ell_P^3$. **[B]**

$S_{\text{BH}} = A/(4\ell_P^2) = V_{\text{shell}}/(4\ell_P^3)$; Bekbits as base unit, $\Omega = 2^b$.

FFGFT

Minimal information unit: winding quantum $\Delta w = 1$ on $\pi_1(T^4) \cong \mathbb{Z}^4$. Discreteness is topological (no half winding), not from a tiling (Doc. 302).

Planck voxels exist implicitly as orbifold fixed points. Bekenstein: $S_{\text{BH}} = A/(4\ell_{\text{P}}^2)$ as a consequence of the Bekenstein bound on the saturation surface **[B]**.

Holographic direction: N -dimensional $\rightarrow$ 3D (as Matzke), not $3D \rightarrow 2D$.

Convergence: fermion generation = algebraic base unit

Matzke: $\text{Cl}(6) = 3$ Witt pairs = Planck voxel = one fermion generation (Furey). **[B]**

FFGFT: $T^4/\mathbb{Z}_3$ orbifold, three $\mathbb{Z}_3$ sectors $\rightarrow \text{SU}(3)_c \rightarrow N_c = 3$ algebraically necessary $\rightarrow$ one fermion generation. **[B]** (Doc. 321)

Both find: threefold structure, algebraically enforced, no free parameter.

Divergence: depth of derivation

Matzke presupposes $\text{Cl}(6)$ as physical unit (**[POSTULATE]** via Furey identification).

FFGFT derives the threefold structure from $\mathbb{Z}_3$ trialization: $N_c = 3$ follows with algebraic necessity **[B]** (Doc. 321, R79–R80).

.5 Schwarzschild radius and Bekenstein entropy

Matzke's Novel Result #1:

$$r_s = \frac{2}{\sqrt{\pi}} \sqrt{S_{\text{BH}}} \ell_{\text{P}} \quad (\text{Matzke})$$

The factor $2/\sqrt{\pi}$ comes from the transition hypercube $\rightarrow$ sphere ("rounding the square").

FFGFT:

$$G = \frac{\xi^2}{4m_e}, \quad \ell_{\text{P}} = \sqrt{\frac{\hbar G}{c^3}}, \quad r_s = \frac{2GM}{c^2} \quad (\text{FFGFT, Doc. 180})$$

Bekenstein: consequence of the bound $S \leq 2\pi RE/(\hbar c)$ on the saturation surface **[B]**.

Numerically: $20 M_{\odot} \Rightarrow r_s = 59.1 \text{ km}$ (both exact).

Convergence

Both derive r_s without the GR field equations. Both show: ℓ_{P} is enforced by self-consistency, not set axiomatically. Numerically identical.

Divergence

Matzke's $2/\sqrt{\pi}$ follows from hypertetrahedron geometry. In FFGFT no such form factor appears; $r_s = 2GM/c^2$ is the standard relation, and G comes from ξ .

.6 Stability threshold**Matzke**

$n_{\text{thresh}} = 2\pi/(\sqrt{2} \ln 2) = 6.41$ bits. **[B]**

$\text{Cl}(6)$ fermion: 6.000 bits, 0.41 bits below threshold.

A stable universe, because the minimal fermion algebra lies algebraically below the

collapse threshold.

Dependence: n_{thresh} hinges essentially on $m_{\text{bit}} = m_P \ln 2 / (2\pi)$, i.e. on the Landauer bridge with $T = T_P$.

FFGFT

Stable matter follows from $\tilde{T} \cdot m = 1$: lepton masses from rational invariants (r_i, p_i) of the torus space **[K]** (Docs. 006, 046, 320).

No direct bit-counting threshold; stability from the orbifold fixed-point structure.

FFGFT does not know n_{thresh} as a bit count — and it should be system-dependent if bit values are system-dependent.

Open bridge — a system-dependent threshold?

Matzke's $n_{\text{thresh}} = 6.41$ is elegant but hinges on the postulate $m_{\text{bit}} = m_P \ln 2 / (2\pi)$ (universal bit value, Planck temperature). From the FFGFT standpoint the threshold would be system-dependent:

$$n_{\text{thresh}}(L) = \frac{2\pi}{\sqrt{2}} \cdot \frac{m_P c^2}{E_{\text{bit}}(L) \cdot \sqrt{2}} = \frac{2\pi}{\sqrt{2}} \cdot \frac{m_P \cdot L}{\sqrt{2} \hbar / c} \quad (2)$$

For $L = \ell_P$ (Planck): gives the same number as Matzke. For $L = L_0 = \xi \cdot \ell_P$ (FFGFT floor): n_{thresh} would be smaller by a factor ξ .

Question: Is $n_{\text{thresh}} = 6.41$ the right number for the universe, or is it a consequence of the choice $L = \ell_P$? **[S]**

.7 Hawking radiation

Matzke

Hawking quanta are exclusively massless vacuum states $\{V, V_s\}$ of $\text{Cl}(6)$. Annihilation: $P_k^+ \cdot P_-^{\text{vac}} = 0$ (exact in $\mathbb{Z}_3$). **[B]**

Information conserved: each V encodes the orientation of the destroyed Bekenstein voxel.

$T_H = T_P / (4 n_{\text{mass}} \ln 2)$. **[B]**

No formation radiation (formation = time reversal of annihilation).

FFGFT (Doc. 325)

Complete mechanism from FFGFT-native building blocks:

Temperature: KMS rule $T_H = \hbar / (k_B \tau)$ with $\tau = 8\pi G M / c^3$ from $\tilde{T} \cdot m = 1$ **[K]** (Docs. 313 G, 325).

Quantum: $k_B T_H = \hbar c / (4\pi r_s)$ exactly — the Hawking quantum is the system-dependent bit of the horizon scale **[K]** (Doc. 325).

Selection: only $T_R = 0$ escapes; absorption $P_j P_k = 0$; confinement = Hawking selection **[B]** (Docs. 321, 325).

Carrier: massless $n_4 = 0$ torus mode; massive modes fall back (clock compression) **[K]** (Doc. 325).

Information: sector pair $(k, -k)$ orthogonally readable, unitarity of the spectral operator **[K]** (Docs. 322, 325).

Convergence — two complete mechanisms

Both frameworks possess an algebraically complete mechanism. The correspondences are structurally exact (Doc. 325): Matzke's $P_k^+ \cdot V_s = 0 \leftrightarrow$ FFGFT's $P_j P_k = 0$; Matzke's massless $V \leftrightarrow$ FFGFT's $n_4 = 0$ torus mode; Matzke's voxel orientation $\leftrightarrow$ FFGFT's sector pair $(k, -k)$.

Divergence — the bit-value foundation

Matzke's mechanism rests on the universal bit value $m_{\text{bit}} = m_P \ln 2 / (2\pi)$; the FFGFT route (KMS from $\tilde{T} \cdot m = 1$ plus system-dependent bit energy) does without it and shows: it is a reparametrization of the KMS relation, not an independent physical input (Doc. 325). In addition FFGFT delivers the power correction $P = P_{\text{std}}(1 - \xi \ln(M/m_P)) = 0.6\text{--}1.4\%$, for which Matzke has no counterpart.

.8 Comparison table

Quantity	Matzke/hyperbit	FFGFT
Fundamental object	$e_i^2 = +1$ [POSTULATE]	$\xi = 4/30000$ from spectrum [K]
Bit energy	$E_{\text{bit}} = k_B T_P \ln 2$ (universal) [POSTULATE]	$E_{\text{bit}} = \hbar c / L$ (system-dependent) [K]
Role of Landauer	physical bridge: $m_{\text{bit}} = m_P \ln 2 / (2\pi)$	carrier operation, not abstract info (A271)
Bit value	universal, $\approx 0.11 m_P$	system-dependent, scale- L -dependent
Planck length	Compton Schwarzschild [B]	$= G = \xi^2 / (4m_e)$, $\ell_P = \sqrt{\hbar G / c^3}$ [K]
Minimal length	ℓ_P	$L_0 = \xi \cdot \ell_P \approx \ell_P / 7500$ [K]
Stability threshold	$n_{\text{thresh}} = 6.41$ bits [B]	orbifold fixed-point structure, no bit count
r_s without GR	$(2/\sqrt{\pi})\sqrt{S_{\text{BH}}} \ell_P$ [B]	$2GM/c^2$ with G from ξ [K]
α^{-1}	not derived	137.036 from $D_f = 3 - \xi$ [K]
Lepton masses	not derived	m_e, m_μ, m_τ from (r_i, p_i) [K]
$\sin^2 \theta_W$	not derived	0.2308 [K] (Doc. 323)
Hawking mechanism	V/V_s annihilation [B]	KMS + $\mathbb{Z}_3$ selection [K]/[B] (Doc. 325)
$N_c = 3$ algebraically	Furey identification [POSTULATE]	$\mathbb{Z}_3$ trialization [B] (Doc. 321)
Holography	$N\text{-dim} \rightarrow 3D$ [B]	$T^4/\mathbb{Z}_3$ projection [K]
Information conservation	V encoding of voxel orientation [B]	unitarity of the spectral operator [K]

.9 FFGFT objections to Matzke's Landauer bridge

Three concrete objections from the FFGFT corpus:

1. **Landauer applies to carrier operations, not to abstract information** (A272, A273). What Landauer 1961 proves is a lower bound for *carrier operations on a thermal equilibrium ensemble* — not a universal statement about abstract information. The identification $m_{\text{bit}} = m_p \ln 2 / (2\pi)$ treats the bit as a thermodynamic object at Planck temperature. That is a realization assumption, not a law.
2. **The bit energy is scale-specific** (Docs. 302, 257). In FFGFT the minimal information unit is the winding quantum $\Delta w = 1$ on T^4 — scale-universal as a *counting quantity*, but with scale-specific energy $E_{\text{bit}} = \hbar c / L$. Landauer's $k_B T \ln 2$ is the special case at one particular scale and bath temperature. The phrase “the bit energy is $m_p \ln 2 / (2\pi)$ ” implies a scale that is not distinguished in FFGFT.
3. **The stability threshold is not a purely algebraic statement** (derived from points 1 and 2). $n_{\text{thresh}} = 2\pi / (\sqrt{2} \ln 2) = 6.41$ follows from the Compton–Schwarzschild equation plus $m_{\text{bit}} = m_p \ln 2 / (2\pi)$. The first piece (Compton = Schwarzschild) is geometric **[B]**. The second (Landauer at T_p) is a postulate. The apparent elegance of the result (Cl(6) exactly 0.41 bits below threshold) hinges on this postulate. A system-dependent bit energy would shift the threshold.

.10 Open bridges

Points with potential for mutual cross-fertilization

- [S]** n_{thresh} **from the orbifold structure:** Doc. 325 shows that Matzke's bit value is a reparametrization of the KMS relation. The constructive counter-question remains open: can the FFGFT fixed-point condition for fermion stability be translated into an effective bit-counting statement — and which system-specific bit energy then enters?
- [S]** **Casimir connection** (Doc. 324, R84): $\xi = C_2(\text{SU}(3)_{\text{fund}}) / N_{\text{Fourier}} = (4/3) / 10^4$. This links the algebraic structure of Cl(6) (Casimir invariant $C_2 = 4/3$, from $N_c = 3$ **[B]**) with the geometric scale of FFGFT (torus mode number $N_F = 10^4$). This bridge is closed **[K]** (Doc. 324).
- [S]** $\sin^2 \theta_W$ **from Cl(6):** FFGFT derives $\sin^2 \theta_W(M_Z) = 0.2308$ **[K]** (Doc. 323). Can Matzke's Cl(6) structure produce an analogous value?

.11 Status balance

Result	Matzke	FFGFT
r_s without GR	[B]	[K]
ℓ_p from self-consistency	[B]	[K]
$Cl(6)$ = fermion generation	[B] (Furey)	[B] (orbifold)
$N_c = 3$ algebraically necessary	[POSTULATE]	[B]
Bit energy system-dependent	[X]	[K]
$\alpha^{-1} = 137.036$	[X]	[K]
Lepton masses	[X]	[K]
$\sin^2 \theta_W$	[X]	[K]
Hawking mechanism worked out	[B]	[K]/[B] (Doc. 325)
n_{thresh} (bit count, universal)	[B]	[X] (rejected: system-dep.)
Information conservation	[B]	[K]
Holography $\neq 3D \rightarrow 2D$	[B]	[K]

Conclusion

Matzke's hyperbit framework and FFGFT are complementary structures with remarkably similar structural findings: fermion generation = algebraic base unit, $N_c = 3$ enforced, Schwarzschild radius without GR.

The deepest difference lies in the treatment of bit values. Matzke posits a universal bit value via the Landauer bridge at Planck temperature — an elegant but, by FFGFT standards, inadmissible universalization. FFGFT holds: bit values are system-dependent ($E_{\text{bit}} = \hbar c/L$), Landauer applies to carrier operations of a thermal ensemble, and the stability threshold n_{thresh} would then not be a universal number.

This is no objection to the algebraic beauty of Matzke's approach, but a theoretical difference in foundations. Doc. 325 makes it constructive: the complete Hawking physics — temperature, selection, carrier, information conservation — follows from $\tilde{T} \cdot m = 1$, the $\mathbb{Z}_3$ structure and the system-dependent bit energy, without a universal bit value. Matzke's $n_{\text{thresh}} = 6.41$ thus demonstrably hinges on the bit-value parametrization, not on the temperature physics.

Main finding

Matzke and FFGFT converge on **fermion generation = fundamental algebraic unit**. They diverge on **bit values**: universal (Matzke, Landauer at T_p) vs. system-dependent (FFGFT, $E_{\text{bit}} = \hbar c/L$). This difference is not cosmetic — it concerns the stability threshold and the mass of a bit. The Hawking temperature formula itself is untouched by it: both routes give the same number, but only the FFGFT route (Doc. 325) does without a universal bit value.

Relation to the FFGFT corpus

Topic	Status	Doc.
Landauer: phase-space bookkeeping	[K]	A271
Carrier vs. information (Landauer critique)	[K]	A272
Bit energy system-dependent (computing sphere)	[K]	A273
Information unit $\Delta w = 1$, $E_{\text{bit}} = \hbar c/L$	[K]	Doc. 257
Elementary cells, bit capacity, partition invariance	[K]	Doc. 302
$G = \xi^2/(4m_e)$, ℓ_P , L_0	[K]	Doc. 180
$SU(3)_c$ from $\mathbb{Z}_3$ trialization	[B]	Doc. 321
$\sin^2 \theta_W(M_Z) = 0.2308$	[K]	Doc. 323
$\xi = C_2(SU(3))/N_{\text{Fourier}}$	[K]	Doc. 324, R84
Hawking mechanism (KMS, $\mathbb{Z}_3$, $n_4=0$)	[K]/[B]	Doc. 325
Spectral theory, Hilbert-space embedding	[K]	Doc. 322
Lepton masses from (r_i, p_i)	[K]	Docs. 006, 046, 320
$E = mc^2 = E = m$ (unit identity)	[K]	Doc. 077

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